

Final Report

Project Title

Design and Simulation of a Virtual Enterprise Network Using CISCO packet tracker and Wireshark

Student Name: Temitayo Oluwaseun

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1. Project Overview

The purpose of this project was to design, configure, test, and document a small virtual computer network. The network was built using networking simulation tools to demonstrate how networking devices communicate using common networking protocols. The project included the configuration of routers, switches, client computers, and a server to create a functional client-server network.

2. Objectives of the Project

The objectives of this project were to:

- Design a functional virtual network.
 - Configure network devices.
 - Demonstrate communication between client and server devices.
 - Understand IP addressing and subnetting.
 - Demonstrate networking protocols including DHCP, DNS, ARP, HTTP/HTTPS, ICMP, TCP, UDP, NAT, and SSL/TLS.
 - Analyze network traffic using Wireshark.
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3. Network Design Explanation

The network consists of one router connected to one switch. Two client computers and one server are connected to the switch. The router serves as the gateway between the local network and an external network. The server provides network services while the client computers communicate with the server through the switch.

The network topology includes:

- 1 Router
 - 1 Switch
 - 2 Client PCs
 - 1 Server
-

4. Description of Networking Protocols Used

DHCP

Dynamic Host Configuration Protocol automatically assigns IP addresses to devices joining the network.

DNS

Domain Name System translates host names into IP addresses.

ARP

Address Resolution Protocol resolves IP addresses into MAC addresses for communication on the local network.

HTTP/HTTPS

Used to provide web communication between clients and the server.

ICMP

Used to test connectivity using ping.

TCP

Provides reliable communication between devices.

UDP

Provides fast communication without guaranteed delivery.

NAT

Allows private IP addresses to communicate with external networks.

SSL/TLS

Provides secure encrypted communication.

5. Device Configurations

The network devices included:

- Router
- Switch
- PC1
- PC2
- Server

Example IP Address Table

Device	IP Address	Subnet Mask	Default Gateway
Router	192.168.1.1	255.255.255.0	N/A
PC1	192.168.1.2	255.255.255.0	192.168.1.1
PC2	192.168.1.3	255.255.255.0	192.168.1.1
Server	192.168.1.4	255.255.255.0	192.168.1.1

6. Network Testing and Results

The network was tested using:

- Ping
- Trace-route

- Wireshark packet capture
- HTTP/HTTPS testing
- DHCP testing
- DNS testing

The successful ping results confirmed communication between devices. Wireshark captured ARP, ICMP, TCP, UDP, DNS, and HTTP packets, demonstrating proper protocol operation.

7. Challenges Faced and Solutions

During the project, several challenges were encountered.

- Packet Tracer experienced issues with static IP configuration on Windows 11.
- Network configuration required troubleshooting to establish successful communication.
- Software compatibility issues required reinstalling networking tools.

These problems were resolved through troubleshooting, software reinstallation, and testing different configuration methods.

8. Conclusion

This project provided practical experience in designing and configuring a virtual computer network. It improved understanding of networking devices, IP addressing, routing, switching, and common networking protocols. The project also demonstrated the importance of troubleshooting and network testing using Wireshark and other networking tools.

Screenshot

LogicalPhysicalv 801 v 100

03:49:00

PC-PT PC1

PC-PT PC3

2960-24TT Switch0

2911-24TT Switch1

Server-PT Server0

Time: 03:59:16

RealtimeSimulation

Scenario 5

Now

Go to PC0 Last Window

File: Last Status | Source | Destination | Type | Color | Timesecs | Periodic | Num | Edit | Delete

PC-PT PC1

PC-PT PC0

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

A Record

Add

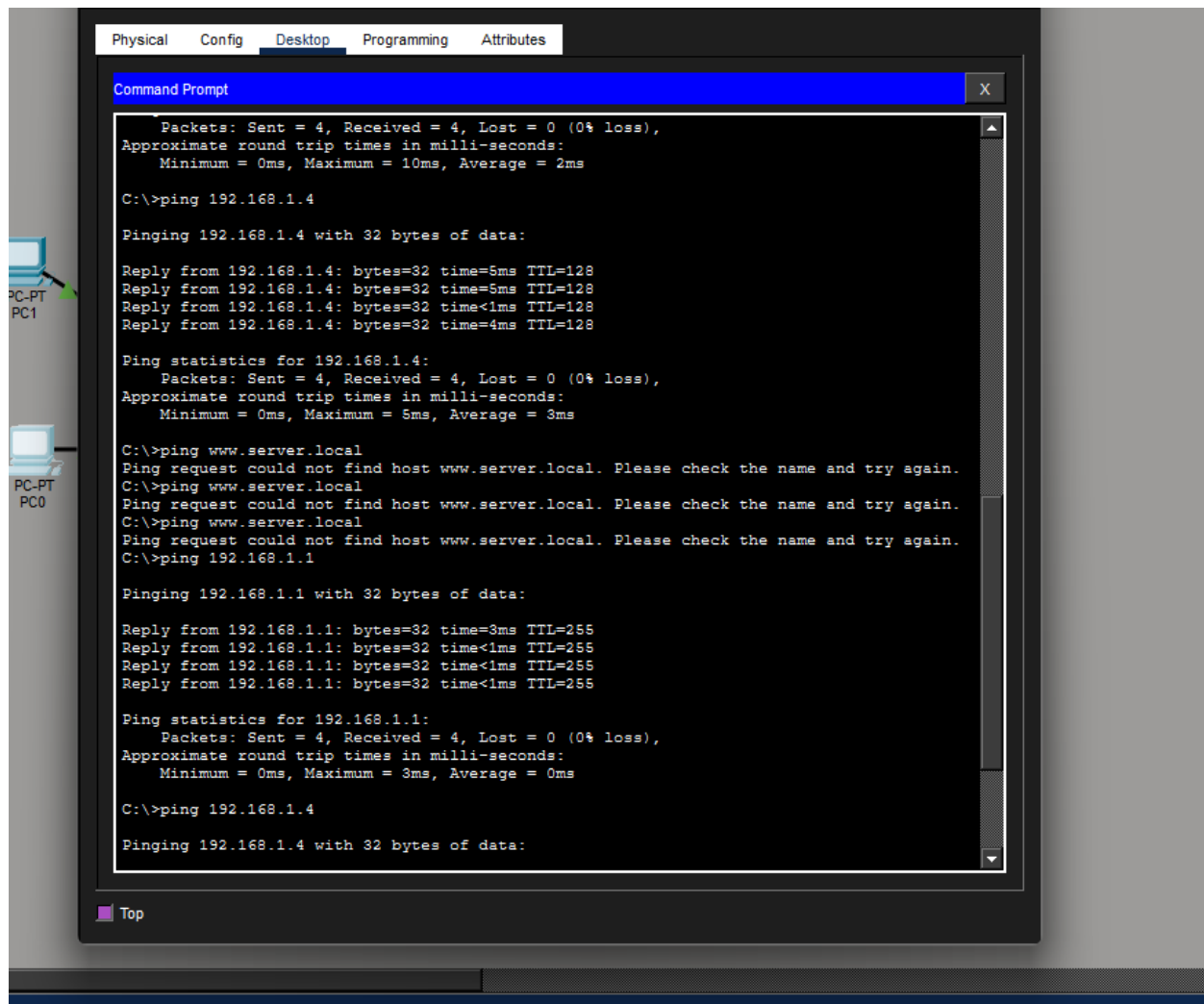
Save

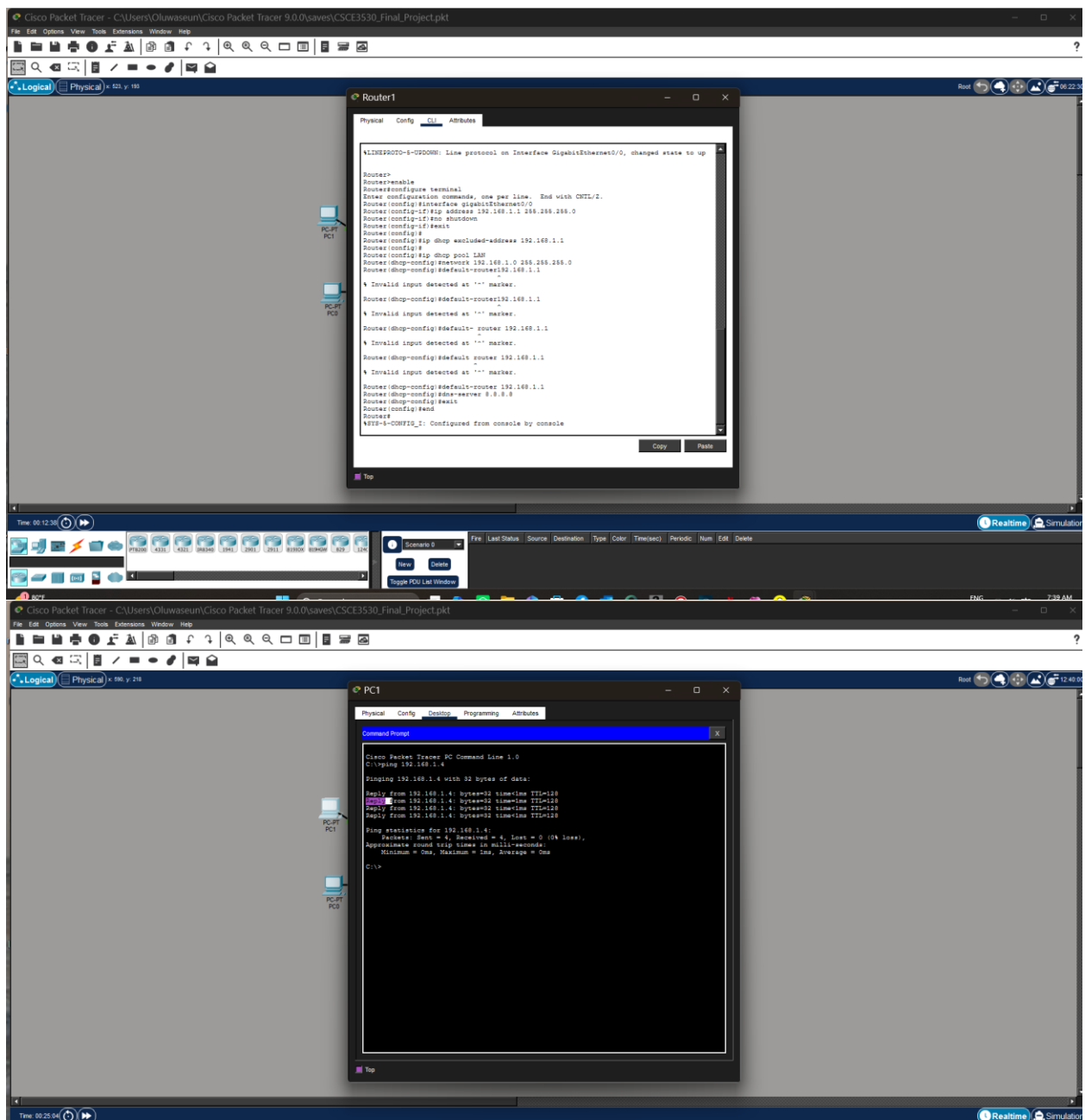
Remove

No.	Name	Type	Detail
0	www.server.local	A Record	192.168.1.4

DNS Cache

Top





Cisco Packet Tracer - C:\Users\Oluwaseun\Cisco Packet Tracer 9.0.0\saves\CSCE3530_FinalProject.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical 900 v 100

PC0

Physical Config Destination Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.4
Pinging 192.168.1.4 with 32 bytes of data:
Reply from 192.168.1.4: bytes=32 time=1ms TTL=255
Reply from 192.168.1.4: bytes=32 time=1ms TTL=255
Reply from 192.168.1.4: bytes=32 time=1ms TTL=255
Reply from 192.168.1.4: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

Time: 00:25:11

Realtime Simulation

*WiFi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

arp

No.	arp	Time	Source	Destination	Info	Protocol	Length
71	4.778709500	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
93	9.736968900	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
120	14.726049000	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
225	19.722201900	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
249	24.749315700	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
641	29.765761600	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
792	34.774854600	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
1028	39.722872300	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
1060	44.735924300	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
1077	49.723823400	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
1100	54.742206600	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
1188	59.769958600	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
1280	64.786914900	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
1299	69.723264800	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
1419	74.725678300	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	
1770	79.731610100	Espressif_1d:b5:84	Broadcast	ARP Announcement for 100.70.19.77	ARP	42	

Frame 71: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device

Ethernet II, Src: Espressif_1d:b5:84 (14:33:5c:1d:b5:84), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Address Resolution Protocol (ARP Announcement)

0000 ff ff ff ff ff 14 33 5c 1d b5 84 08 06 00 013 \.....

0010 08 00 06 04 00 01 14 33 5c 1d b5 84 64 46 13 4d3 \...dF M

0020 00 00 00 00 00 00 64 46 13 4ddF M

Wireshark network traffic analysis interface showing DHCP and TCP packet captures.

Top Panel: DHCP Filter

No.	Time	Source	Destination	Info	Protocol	Length
594	27.614112500	0.0.0.0	255.255.255.255	DHCP Discover - Transaction ID 0x89a14814	DHCP	342

Packet Details (Frame 594):

- Frame 594: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface \Device\NPF...
- Ethernet II, Src: Routerboardc_86:cd:29 (f4:1e:57:86:cd:29), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
- Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
- User Datagram Protocol, Src Port: 68, Dst Port: 67
- Dynamic Host Configuration Protocol (Discover)

Packet Bytes:

Offset	Hex	ASCII
0000	ff ff ff ff ff f4 1e 57 86 cd 29 08 00 45 00	W.....E
0010	01 48 00 00 00 00 10 11 a9 a6 00 00 00 ff ff	..H.....
0020	ff ff 00 44 00 43 01 34 45 da 01 01 06 00 89 a1	D.C.4.E.....
0030	48 14 00 40 80 00 00 00 00 00 00 00 00 00 00	H-@.....
0040	00 00 00 00 00 00 f4 1e 57 86 cd 29 00 00 00 00	W.....
0050	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0060	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0070	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0080	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0090	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00a0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00b0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00c0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00d0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00e0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00f0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0100	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0110	00 00 00 00 00 00 63 82 53 63 35 01 01 37 06 01	c.Sc5.7.....
0120	03 06 0f 2c 21 3d 07 01 f4 1e 57 86 cd 29 ff 00	...[...W.....
0130	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

Dynamic Host Configuration Protocol: Protocol

Packets: 2772 - Displayed: 1 (0.0%) - Dropped: 0 (0.0%) Profile: Default

Bottom Panel: TCP Filter

No.	Time	Source	Destination	Info	Protocol
584	27.181922900	104.26.13.49	100.70.19.126	Application Data	TLSv1.3
585	27.181924300	104.26.13.49	100.70.19.126	Application Data	TLSv1.3
586	27.181933500	104.26.13.49	100.70.19.126	Application Data	TLSv1.3
587	27.181946700	104.26.13.49	100.70.19.126	Application Data	TLSv1.3
588	27.181950600	104.26.13.49	100.70.19.126	Application Data, Application Data, Application Data	TLSv1.3
589	27.182096000	100.70.19.126	104.26.13.49	53520 → 443 [ACK] Seq=989 Ack=19545 Win=65280 Len=0	TCP
590	27.183197200	100.70.19.126	104.26.13.49	53520 → 443 [RST, ACK] Seq=989 Ack=19545 Win=0 Len=0	TCP
591	27.264650200	100.70.19.126	104.208.203.89	49408 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1	TCP
592	27.295600200	104.208.203.89	100.70.19.126	443 → 49408 [ACK] Seq=1 Ack=2 Win=7688 Len=0 SLE=1 SRE=2	TCP
593	27.346831700	66.253.214.16	100.70.19.126	[TCP Retransmission] 53 → 50761 [FIN, ACK] Seq=108 Ack=52 Win=64768 Len=0	TCP
598	28.247259600	100.70.19.126	57.144.200.145	Application Data	TLSv1.2
599	28.247259800	100.70.19.126	57.144.200.145	Application Data	TLSv1.2
600	28.268965100	57.144.200.145	100.70.19.126	Application Data	TLSv1.2
601	28.285168800	57.144.200.145	100.70.19.126	Application Data	TLSv1.2
602	28.309484600	100.70.19.126	57.144.200.145	63206 → 443 [ACK] Seq=94 Ack=76 Win=65535 Len=0	TCP
603	28.342942100	100.70.19.126	57.144.200.145	56743 → 443 [ACK] Seq=88 Ack=76 Win=65535 Len=0	TCP
604	29.009732500	66.253.214.16	100.70.19.126	[TCP Retransmission] 53 → 50761 [FIN, ACK] Seq=108 Ack=52 Win=64768 Len=0	TCP
605	29.590595700	100.70.19.126	66.253.214.16	61326 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	TCP

Packet Details (Frame 593):

- Frame 593: Packet, 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface \Device\NPF...
- Ethernet II, Src: Routerboardc_86:cd:29 (f4:1e:57:86:cd:29), Dst: Intel_e1:4a:4c (84:7b:57:e1:4a)
- Internet Protocol Version 4, Src: 66.253.214.16, Dst: 100.70.19.126
- Transmission Control Protocol, Src Port: 53, Dst Port: 50761, Seq: 108, Ack: 52, Len: 0

Wireshark network traffic capture window showing UDP packets. The interface includes a menu bar (File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help) and a toolbar. The packet list pane displays a table of captured packets, with packet 557 selected. The packet details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Data (34 bytes). The packet bytes pane displays the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Info	Protocol
540	26.703881200	149.36.49.100	100.70.19.126	5001 → 65524 Len=27	UDP
542	26.743259600	217.79.118.27	100.70.19.126	5001 → 65525 Len=108	UDP
543	26.743262100	217.79.118.27	100.70.19.126	5001 → 65525 Len=84	UDP
544	26.743620800	100.70.19.126	217.79.118.27	65525 → 5001 Len=45	UDP
546	26.744503500	217.79.118.27	100.70.19.126	5001 → 65525 Len=58	UDP
547	26.744839500	100.70.19.126	217.79.118.27	65525 → 5001 Len=38	UDP
549	26.745557100	100.70.19.126	149.36.49.100	65524 → 5001 Len=77	UDP
551	26.801025200	217.79.118.27	100.70.19.126	5001 → 65525 Len=27	UDP
552	26.802109100	149.36.49.100	100.70.19.126	5001 → 65524 Len=32	UDP
557	26.830921300	100.70.19.126	149.36.49.100	65524 → 5001 Len=34	UDP
594	27.614112500	0.0.0.0	255.255.255.255	DHCP Discover - Transaction ID 0x89a14814	DHCP
595	27.973546600	100.70.19.31	239.255.255.250	M-SEARCH * HTTP/1.1	SSDP
596	28.074704200	100.70.19.31	239.255.255.250	M-SEARCH * HTTP/1.1	SSDP
597	28.178364800	100.70.19.31	239.255.255.250	M-SEARCH * HTTP/1.1	SSDP
757	32.275490800	100.70.19.26	224.0.0.251	Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question	MDNS
758	32.275500100	fe80::824:e52:2a58::	ff02::fb	Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question	MDNS
759	32.2755001800	100.70.19.26	239.255.255.250	M-SEARCH * HTTP/1.1	SSDP
761	32.348178900	100.70.19.96	224.0.0.251	Standard query response 0x0000 PTR SpZc-26C56A._spotify-connect._tcp.local TXT, cache flush A, cache flush 100...	MDNS

Frame 557: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface \Device...
Ethernet II, Src: Intel_e1:4a:4c (84:7b:57:e1:4a:4c), Dst: Routerboardc_86:cd:29 (f4:1e:57:86:cd:29)
Internet Protocol Version 4, Src: 100.70.19.126, Dst: 149.36.49.100
User Datagram Protocol, Src Port: 65524, Dst Port: 5001
Data (34 bytes)

0000 f4 1e 57 86 cd 29 84 7b 57 e1 4a 4c 08 00 45 00 .W.) { W.JL..E.
0010 00 3e 5b be 40 00 80 11 00 00 64 46 13 7e 95 24 .>[@... ..dF~\$.
0020 31 64 ff f4 13 89 00 2a 3e 88 7e 76 b3 23 e6 d1 1d* >~v #..
0030 29 31 aa 6d 4e 81 a6 94 38 f0 df bc 38 8b f1 ef)1 mN... 8...8..
0040 31 48 91 86 7a d9 b0 08 61 f3 31 a5 1h z... a 1